

1	Define an Operating System. Mention its main functions.												
2	Distinguish between system software and application software												
3	State inter- Process communication (IPC)												
4	State fragmentation. Difference between internal and external fragmentation.												
5	Differentiate between process and thread.												
6	Explain OS structure with microkernel approach.												
7	Differentiate between One-to-One multithreading models and many-to-many multithreading models with diagram												
8	Describe page replacement algorithms (FIFO,LRU,OPTIMAL)with examples												
9	Elaborate Process Life Cycle with diagram												
10	Solve the FCFS CPU Scheduling Algorithm<table><tr><th>Process</th><th>CPU (Burst Time in Millisecond)</th></tr><tr><td>P1</td><td>5</td></tr><tr><td>P2</td><td>24</td></tr><tr><td>P3</td><td>16</td></tr><tr><td>P4</td><td>10</td></tr><tr><td>P5</td><td>3</td></tr></table>Calculate 1) Gantt chart 2)Average waiting time 3)Average Response time 4)Average turnaround time.	Process	CPU (Burst Time in Millisecond)	P1	5	P2	24	P3	16	P4	10	P5	3
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11	Discuss about contiguous Memory Allocation in detail												
12	Explain Segmentation With Paging												
13	Demonstrate the above terms in Details 1. Demand Paging 2. Swapping 3. Paging 4. Disk I/O 5. Disk Cache 6. I/O Buffering												
14	The job are scheduled for execution as follows. solve the problem by using i) SJF <table><tr><th>Process</th><th>Arrival</th><th>Burst Time</th></tr></table>	Process	Arrival	Burst Time									
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15	Describe the Evolution of Operating System in detail																
16	Discuss the structure of an Operating System with a neat diagram.																
17	Describe various Operating System services .																
18	Compare Batch OS, Multiprogrammed OS, and Time-sharing OS .																
19	Define a process and explain the process life cycle with diagram.																
20	Demonstrate the Inter-Process Communication (IPC) mechanisms.																
21	Explain the CPU Scheduling Algorithm write Advantages & Disadvantage.																
22	Identify the Critical Section Problem and its requirements.																
23	Apply the FCFS CPU Scheduling Algorithm <table><tr><td>Process</td><td>CPU (Burst Time in Millisecond)</td></tr><tr><td>P1</td><td>5</td></tr><tr><td>P2</td><td>24</td></tr><tr><td>P3</td><td>16</td></tr><tr><td>P4</td><td>10</td></tr><tr><td>P5</td><td>3</td></tr></table> Calculate 1) Gantt chart 2)Average waiting time 3)Average Response time 4)Average turnaround time.					Process	CPU (Burst Time in Millisecond)	P1	5	P2	24	P3	16	P4	10	P5	3
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24	Demonstrate Deadlock prevention and deadlock recovery techniques .																
25	Justify Segmentation with diagrams.																
26	Identify Virtual Memory and Demand Paging .																
27	Explain Page Replacement Algorithms : FIFO and LRU.																
28	Prepare File allocation methods and free-space management .																
29	Manage Disk scheduling algorithms and disk management.																
30	Explain OS security, Virtual Machines, and Cloud OS concepts.																
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58	Examine directory implementation method and file allocation method (linked, Indexed, contiguous)
59	Explain OS security, Virtual Machines, and Cloud OS concepts
60	Demonstrate Deadlock prevention and deadlock recovery techniques.